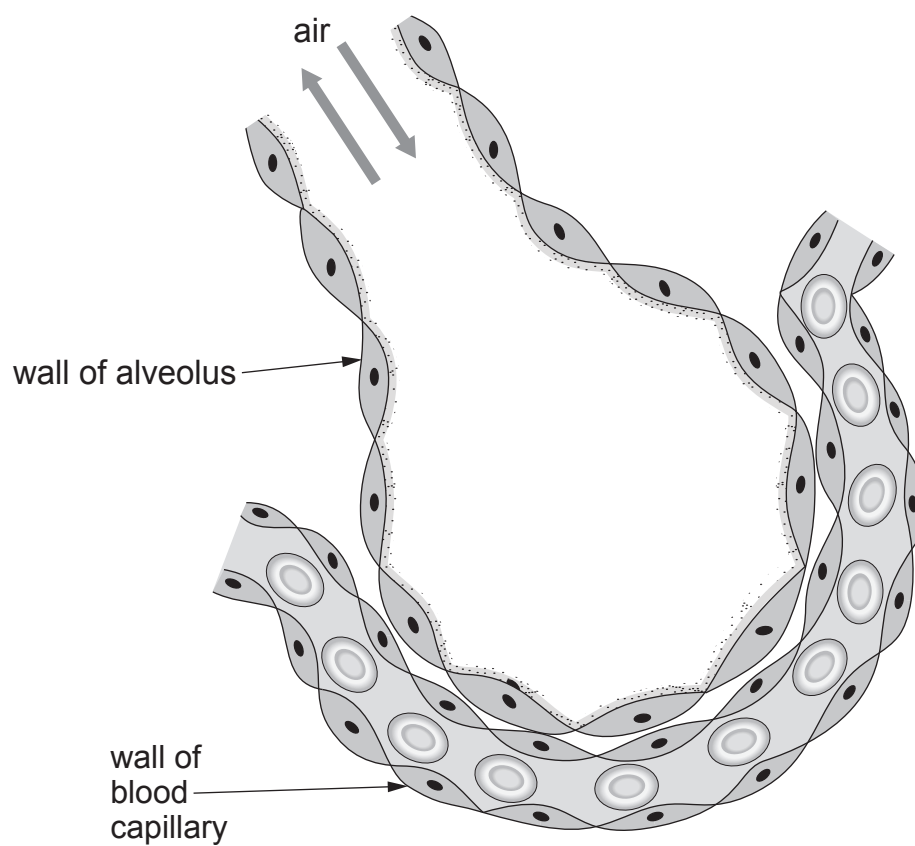


2. **Image 2.1** shows an alveolus.

Image 2.1



(a) **Use labelled arrows** to identify the following structures on **Image 2.1**:

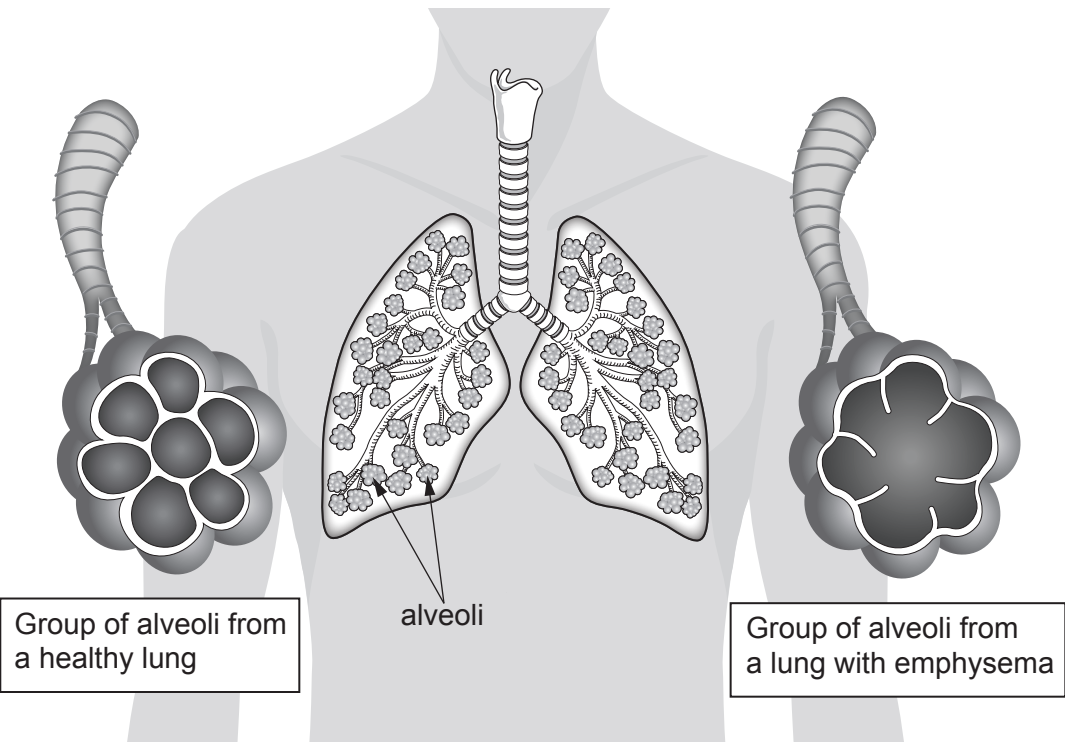
[2]

- (i) bronchiole;
- (ii) blood plasma.



(b) **Image 2.2** shows groups of alveoli from a healthy lung and from a lung of a person with emphysema. The tables below **Image 2.2** show the concentrations of oxygen and carbon dioxide in the blood capillaries.

Image 2.2



Healthy lung	
Gas	Concentration of gases in the blood capillaries (arbitrary units)
oxygen	86
carbon dioxide	45

Lung with emphysema	
Gas	Concentration of gases in the blood capillaries (arbitrary units)
oxygen	53
carbon dioxide	62

Examiner
only

(i) Using **Image 2.2**, explain the differences in the concentrations of gases in the blood capillaries of a healthy lung and a lung with emphysema. [2]

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(ii) State the effect on breathing of the difference in concentrations of these gases for a person suffering from emphysema. [1]

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(iii) State **one** cause of emphysema. [1]

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